

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EC-3004-CBGS

B.E. III Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Electronic Devices and Circuits

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Discuss the principle of Hall effect and its applications.
b) Differentiate between:
 - i) Drift current and diffusion current
 - ii) Transition and diffusion capacitance
2. a) Explain the working of BJT as an amplifier.
b) Discuss about Ebers Moll model.
3. a) Discuss the construction and basic operation of bipolar junction transistor.
b) Explain early effect and base width modulation with diagram.
4. Discuss the selection of operating point, bias stabilization and thermal runaway in respect to transistor.
5. Discuss the small signal analysis of an amplifier using hybrid model.

EC-3004-CBGS

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[2]

6. Discuss the working of following power amplifiers.
 - i) Class A
 - ii) Class AB
7. Explain the construction working and applications of DIAC and TRIAC.
8. Write short notes on :
 - a) Boot strapping techniques
 - b) UJT
